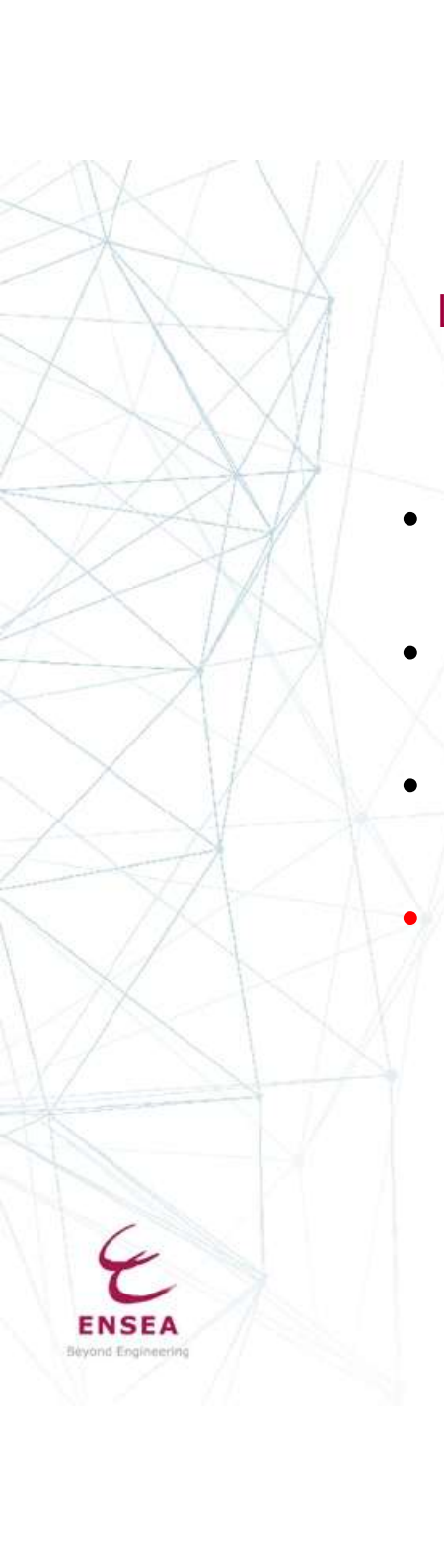




Energy Conversion I

alexis.martin@ensea.fr
Desk D216





Layout

- Non isolated choppers
- Switch mode power supplies
- Power components
- **Sinusoidal absorption**

Introduction

Boost

Flyback

**Example of
converter**

Introduction

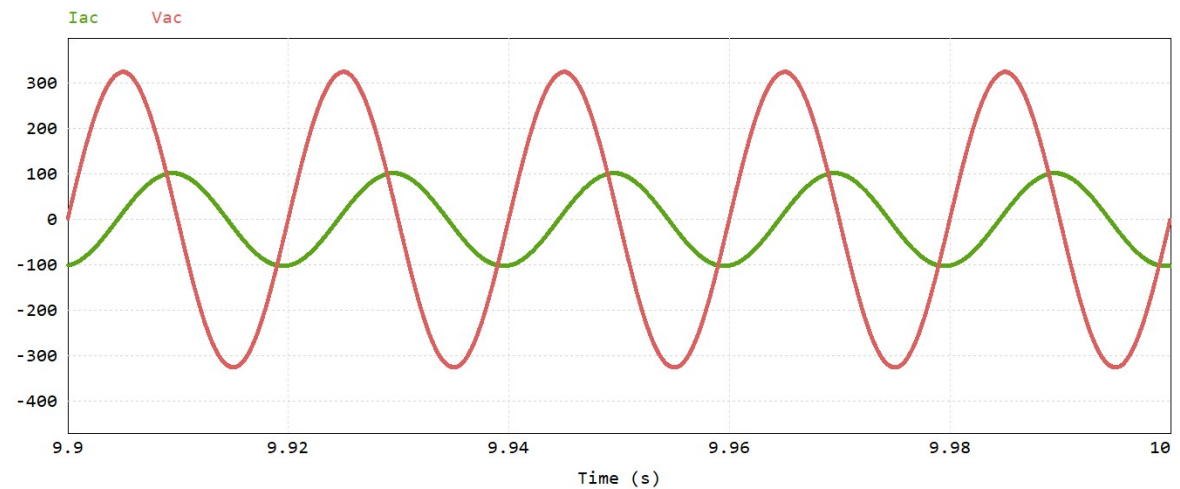
Introduction

Boost

Flyback

Reminder: Power in AC

$$V_{RMS} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$



- Real power: $P = \langle p(t) \rangle = \langle v(t) \cdot i(t) \rangle = V_{RMS} \cdot I_{RMS} \cdot \cos(\varphi_1)$
- Reactive power: $Q = V_{RMS} \cdot I_{RMS} \cdot \sin(\varphi_1)$
- Apparent power: $S = V_{RMS} \cdot I_{RMS}$
- Power factor: $pf = \frac{P}{S}$
- Total harmonic distortion: $THD = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n\ RMS}^2}}{I_{1\ RMS}}$

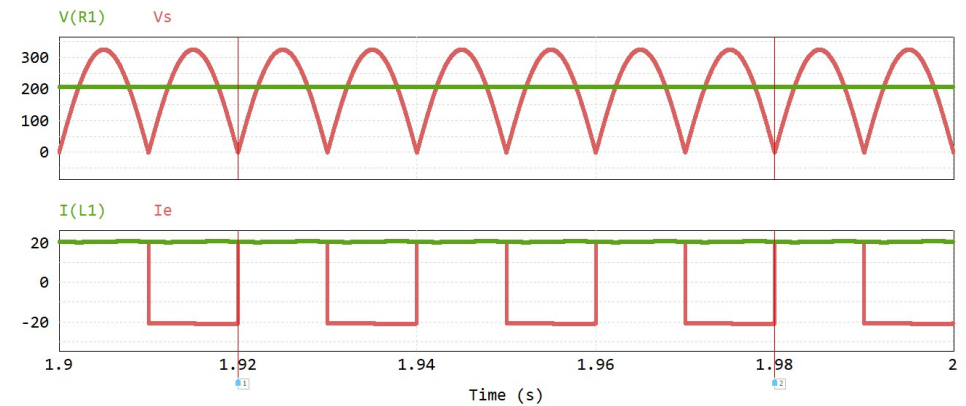
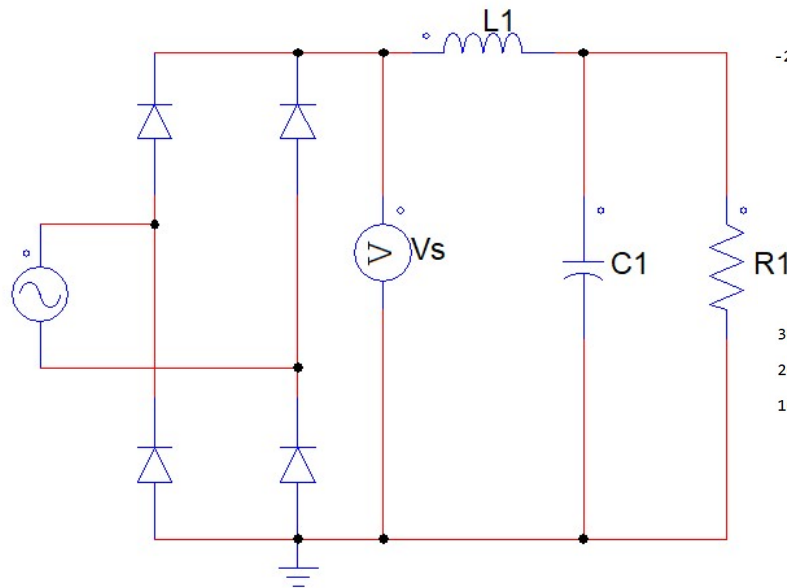
Introduction

Boost

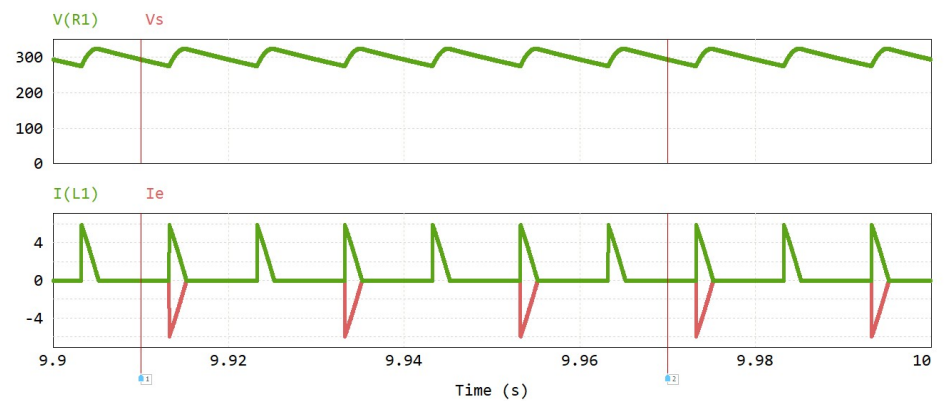
Flyback

Example of
converter

Reminder: Diode rectifier



$$L = 10H \rightarrow pf = 0,9 \quad THD = 0,48$$



$$L = 10nH \rightarrow pf = 0,5 \quad THD = 1,5$$

Introduction

Boost

Flyback

Example of
converter

European standard: 61000-3-2

Limits for harmonics currents emission (class C $\approx$ lightning)

Tableau 2 – Limites pour les appareils de classe C

Rang harmonique n	Courant harmonique maximal exprimé en pourcentage du courant fondamental d'entrée des luminaires %
2	2
3	30 σ
5	10
7	7
9	5
11 δ n δ 39 (harmoniques impairs seulement)	3
* σ est le facteur de puissance du circuit	

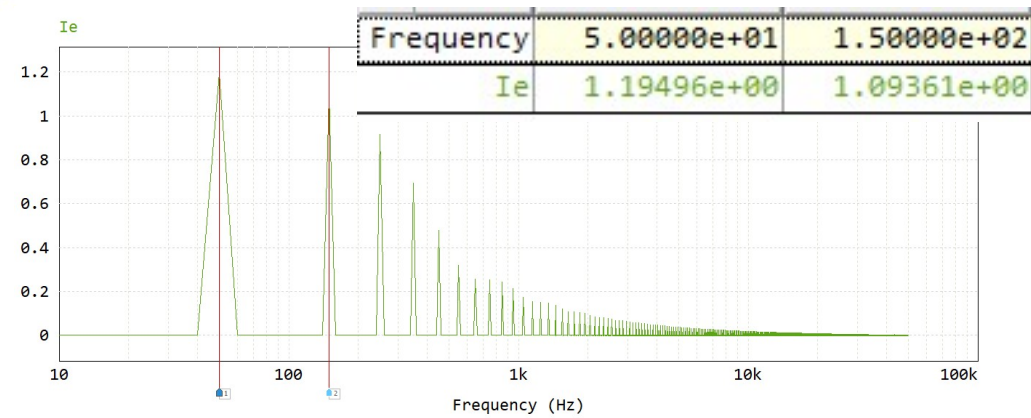
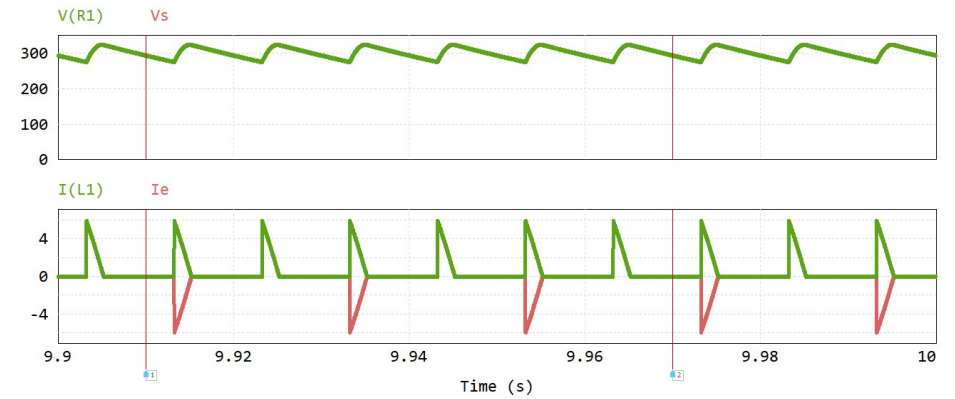
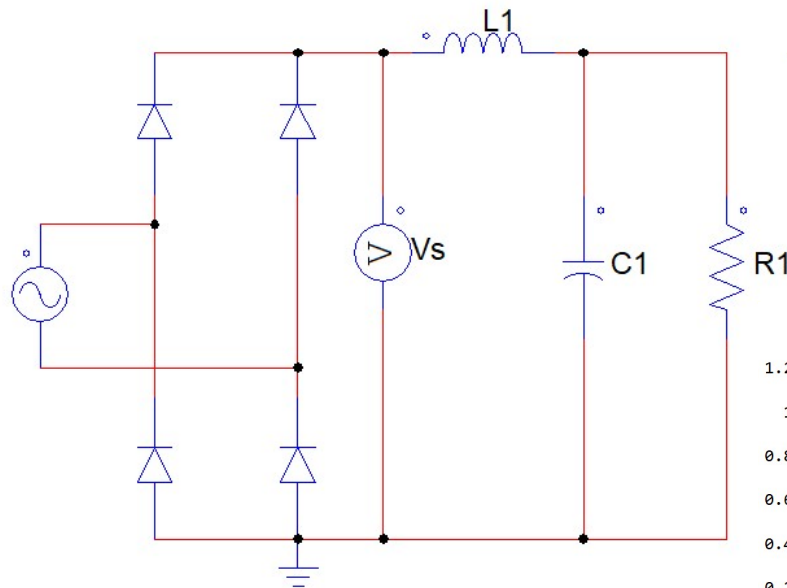
Introduction

Boost

Flyback

Example of
converter

European standard: 61000-3-2



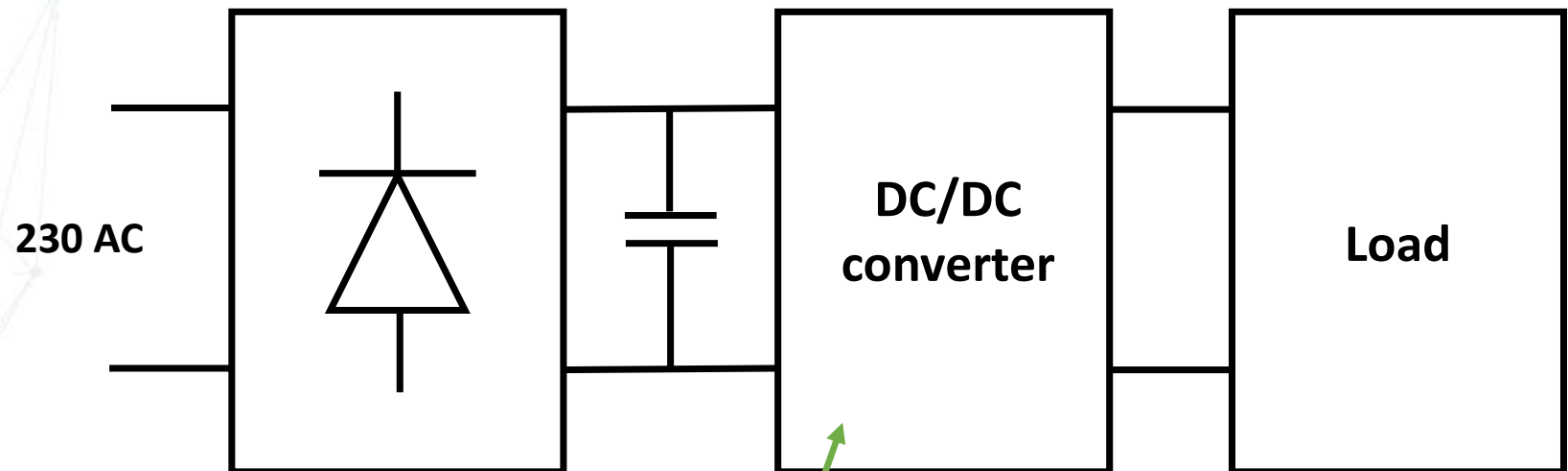
Introduction

Boost

Flyback

Example of
converter

Power factor correction



Used to absorb
sinus current
(before rectifier)

Introduction

Boost

Flyback

Example of
converter

Boost for PFC

Introduction

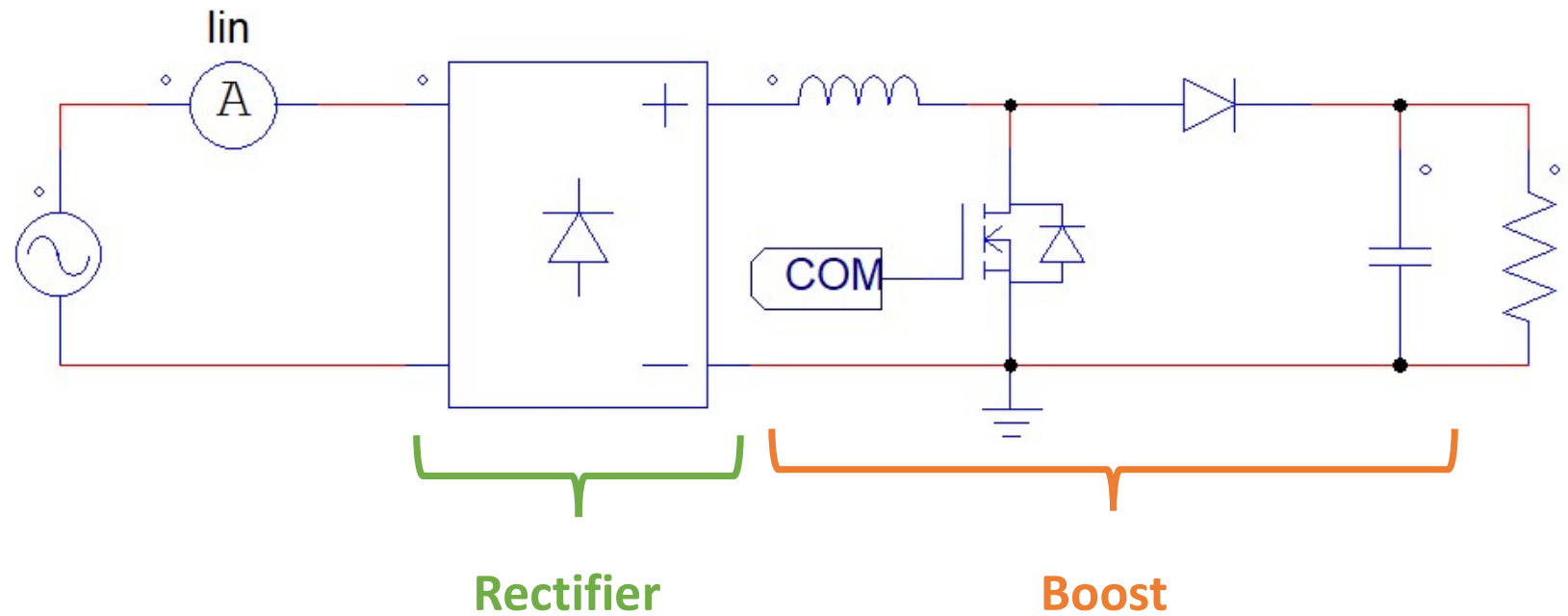
Boost Structure

PWM control
Hysteresis control

Flyback

Example of converter

Structure



Introduction

Boost

Structure

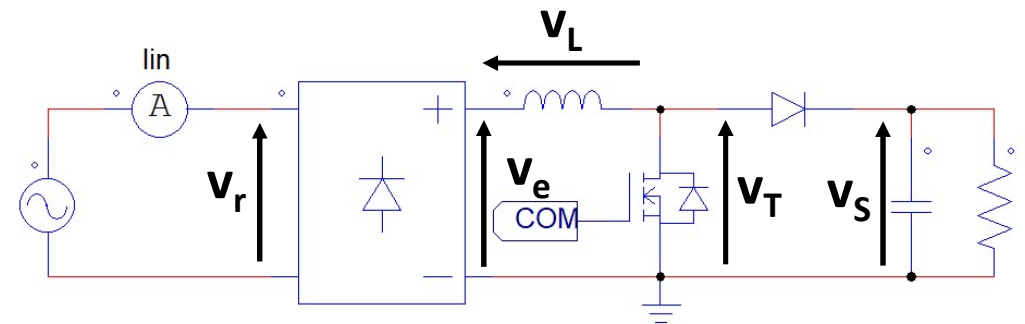
PWM control

Hysteresis control

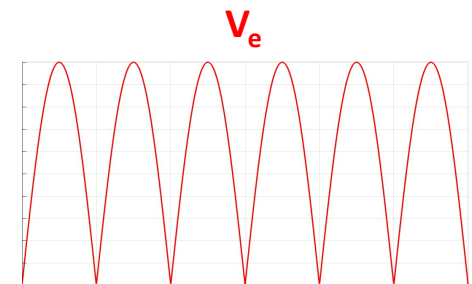
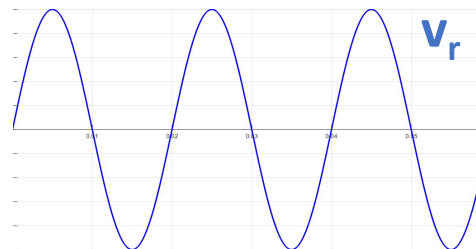
Flyback

Example of converter

PWM control



- With PWM control, the switching frequency is fixed, the duty cycle is used to control the current
- For low frequency, the boost is a continuous voltage amplifier controlled with the duty cycle α
- Continuous conduction of diodes in the rectifier:
$$v_e = |v_r| = V\sqrt{2} \cdot |\sin(\theta)|$$



Introduction

Boost

Structure

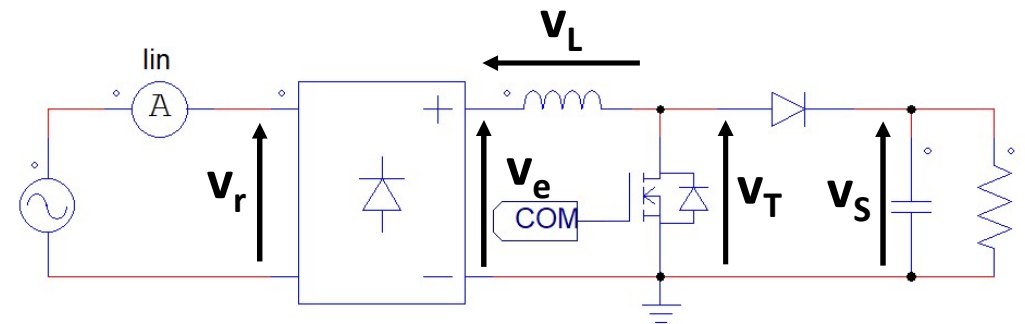
PWM control

Hysteresis control

Flyback

Example of converter

PWM control



- $v_L = v_e - v_T = V\sqrt{2} \cdot |\sin(\theta)| - (1 - \alpha)V_S = L \frac{di_L}{dt}$
- We want $I_{in} = I \cdot \sqrt{2} \sin(\theta)$
 - $i_L = I \cdot \sqrt{2} \cdot |\sin(\theta)|$
- For $0 < \theta < \pi$:
 - $1 - \alpha = \frac{V\sqrt{2}}{V_S} \cdot \sin(\theta) - \frac{L \cdot 2\pi f_{LF} \cdot I\sqrt{2}}{V_S} \cos(\theta)$
 - $\alpha = 1 - \frac{V\sqrt{2}}{V_S} \cdot \left(\sin(\theta) - \frac{2\pi L f_{LF} \cdot P}{V^2} \cos(\theta) \right)$

Introduction

Boost

Structure

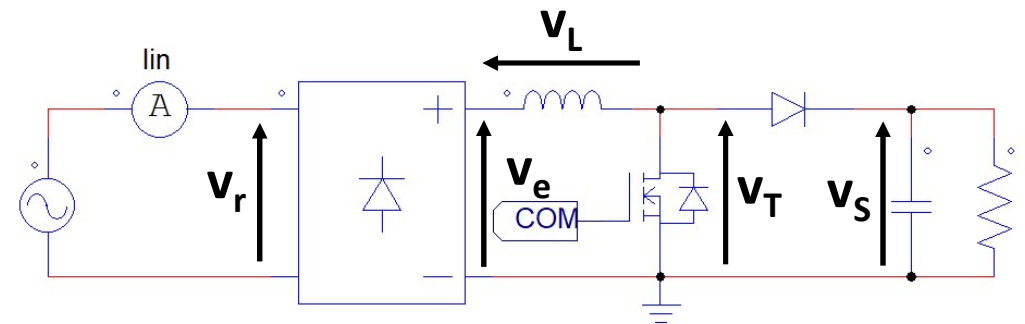
PWM control

Hysteresis control

Flyback

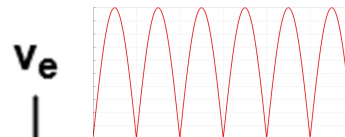
Example of converter

PWM control

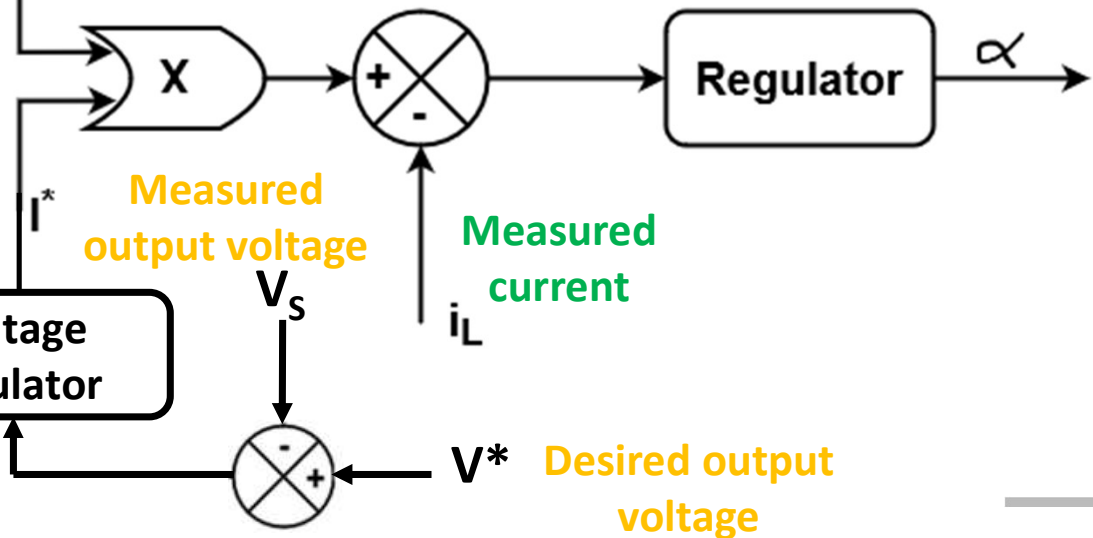


$$\alpha = 1 - \frac{V\sqrt{2}}{V_S} \cdot \left(\sin(\theta) - \frac{2\pi L f_{LF} P}{V^2} \cos(\theta) \right)$$

Input rectified voltage (for synchronization)



Amplitude of the desired current



Introduction

Boost

Structure

PWM control

Hysteresis control

Flyback

Example of converter

PWM control

Input rectified voltage (for synchronization)

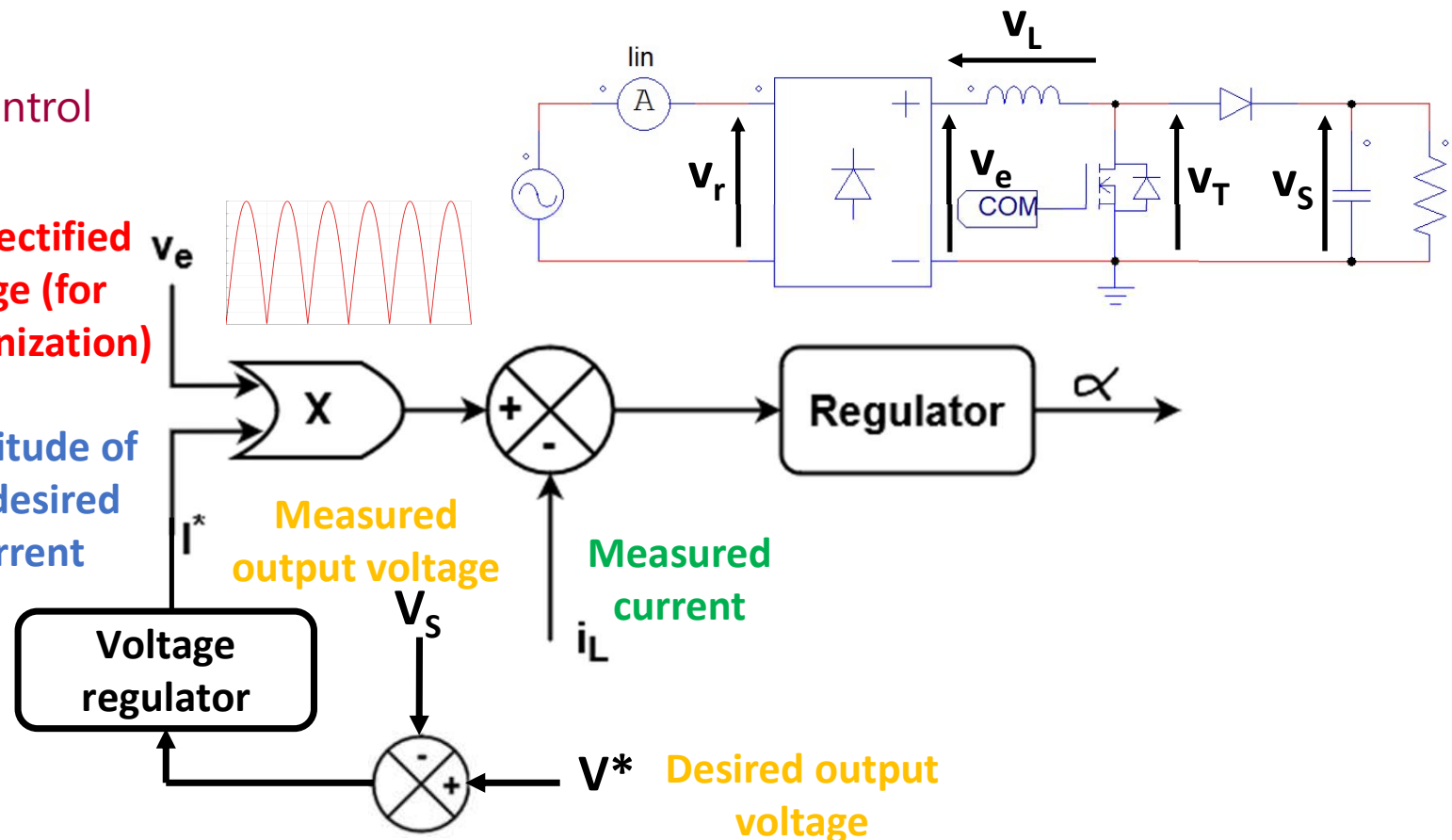
Amplitude of the desired current

Voltage regulator

Measured output voltage

Measured current

Desired output voltage



Condition: voltage control « slow » (few Hz) to ensure sinusoidal current

Filter the output voltage for measurement (sample at $2f_{BF}$, band-stop filter, high output capacitor)

Introduction

Boost

Structure

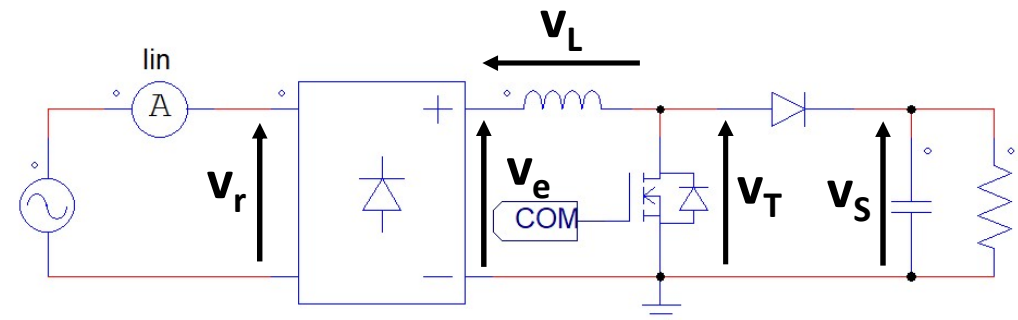
PWM control

Hysteresis control

Flyback

Example of converter

PWM control: components sizing



- Semi-conductors

$$\langle i_D \rangle = \frac{P}{V_S} \quad \langle i_T \rangle = \langle i_L \rangle - \langle i_D \rangle = \frac{P}{V_S} \left(\frac{4}{\pi} \frac{V_S}{V\sqrt{2}} - 1 \right)$$

$$I_{T\text{ RMS}} = \frac{P}{V} \sqrt{1 - \frac{8.V\sqrt{2}}{3\pi V_S}} I_{D\text{ RMS}} = \frac{P}{V} \sqrt{\frac{8.V\sqrt{2}}{3\pi V_S}}$$

- Output capacitor

$$\Delta V_{S\text{ BF}} = \frac{P}{2\pi f_{BF} C V_S} \quad I_{C\text{ RMS}} = \frac{P}{V_S} \sqrt{\frac{16.V_S}{3\pi V\sqrt{2}} - 1}$$

- Input inductor: limit the current ripple due to switching BUT small voltage drop

$$\Delta i_{HF} = \frac{V_S}{4L f_{HF}} \quad \frac{V}{2\pi I f_{BF}} \gg L > \frac{V_S}{4f_{HF} \Delta i_{HF}} \quad I_{L\text{ RMS}} = \frac{P}{V}$$

Introduction

Boost

Structure

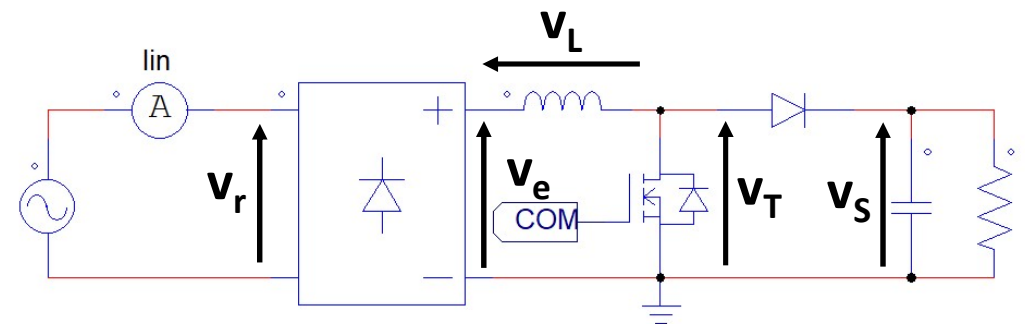
PWM control

Hysteresis control

Flyback

Example of
converter

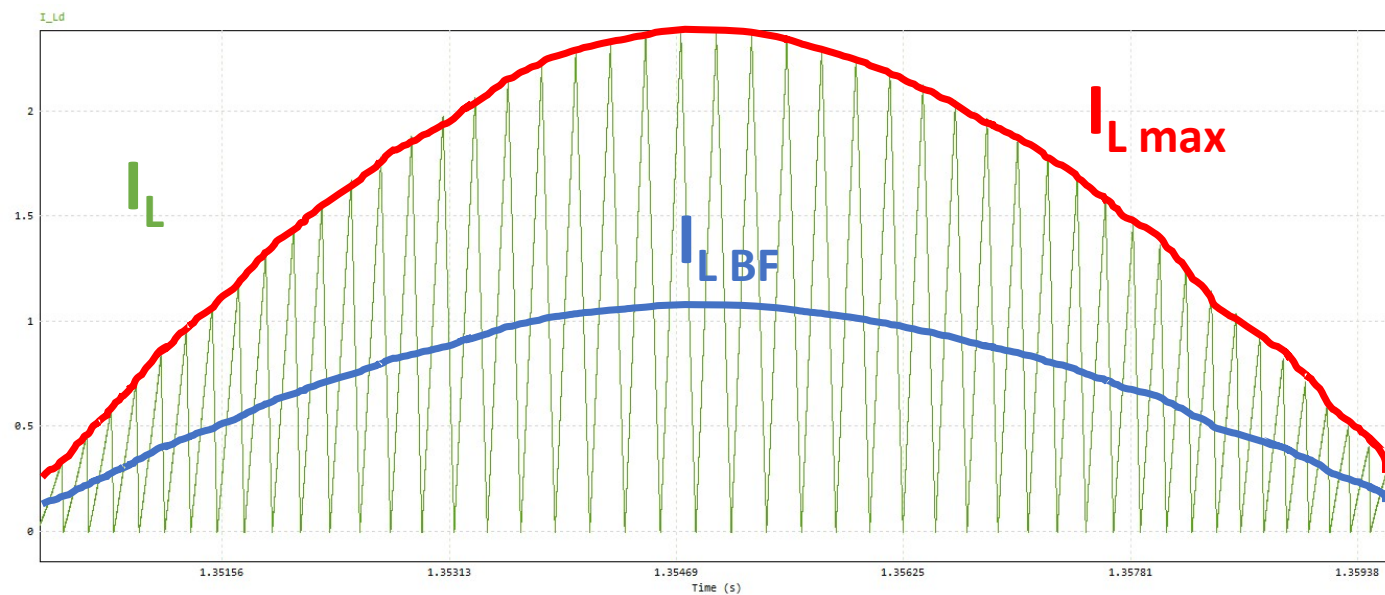
Hysteresis control



PWM control: need a current regulator to follow a sinus input

➤ Complicated to design the regulator

➤ Hysteresis control



Introduction

Boost

Structure

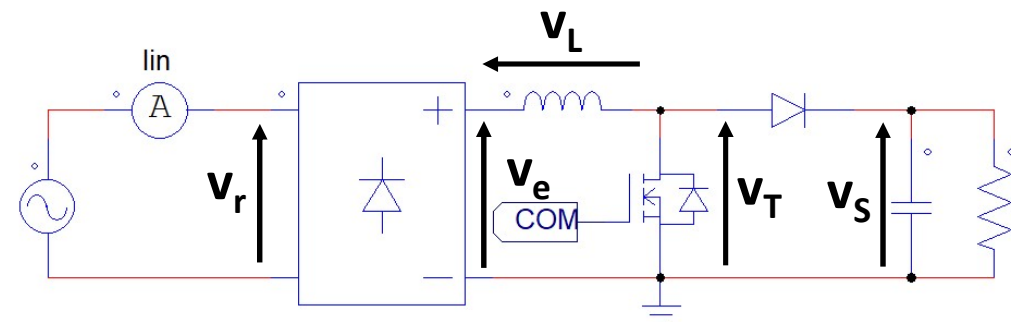
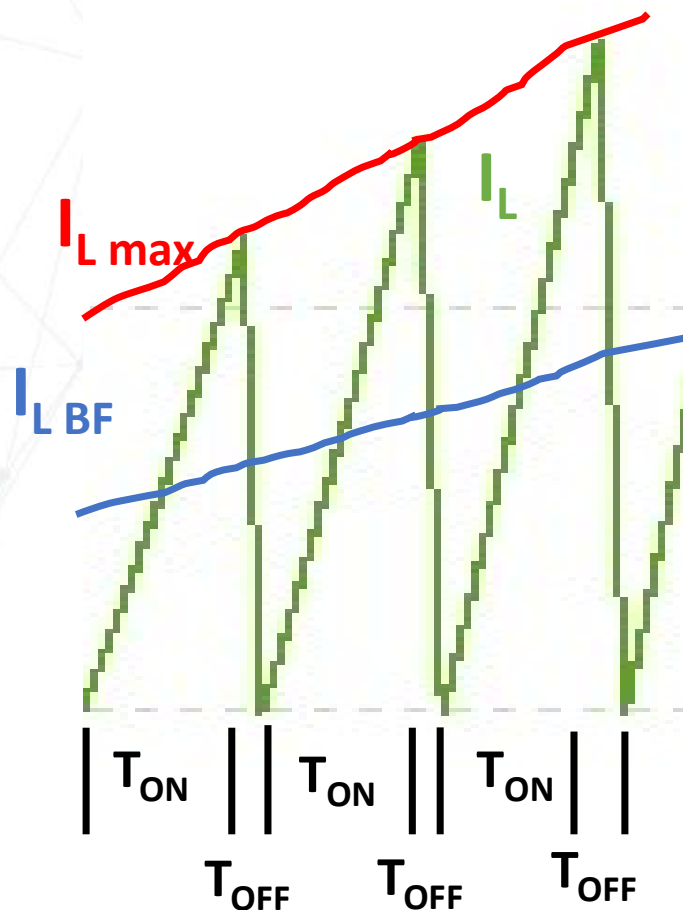
PWM control

Hysteresis control

Flyback

Example of
converter

Hysteresis control



- $I_{L \max} = \frac{V_e}{L} t_{ON} = 2I_{L \text{ BF}} = 2I\sqrt{2}|\sin(\theta)|$
 $\triangleright t_{ON} = \frac{2LI}{V} = \frac{2LP}{V^2} = \text{cst}$
- $t_{OFF} = t_{ON} \frac{1-\alpha}{\alpha} = t_{ON} \frac{V_e}{V_S - V_e}$
 $\triangleright f_{HF} = \frac{1}{t_{ON} + t_{OFF}} = \frac{1}{t_{ON}} \left(1 - \frac{V\sqrt{2}}{V_S} |\sin(\theta)|\right)$
- $f_{HF \min} = \frac{1}{t_{ON}}$
- $f_{HF \max} = \frac{1}{t_{ON}} \left(1 - \frac{V\sqrt{2}}{V_S}\right)$

Introduction

Boost

Structure

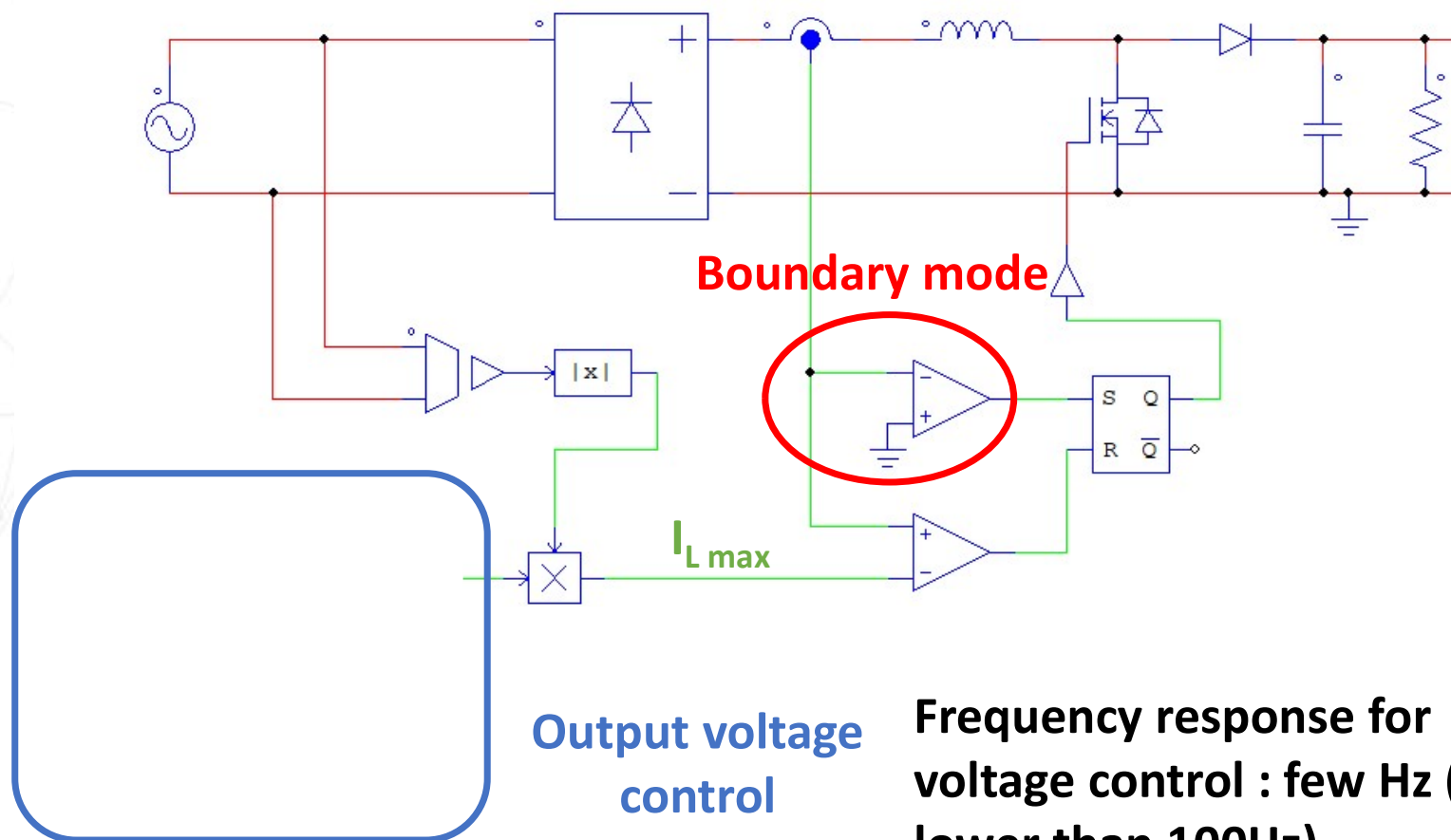
PWM control

Hysteresis control

Flyback

Example of
converter

Hysteresis control



Introduction

Boost

Structure

PWM control

Hysteresis control

Flyback

Example of
converter

Example: LED driver

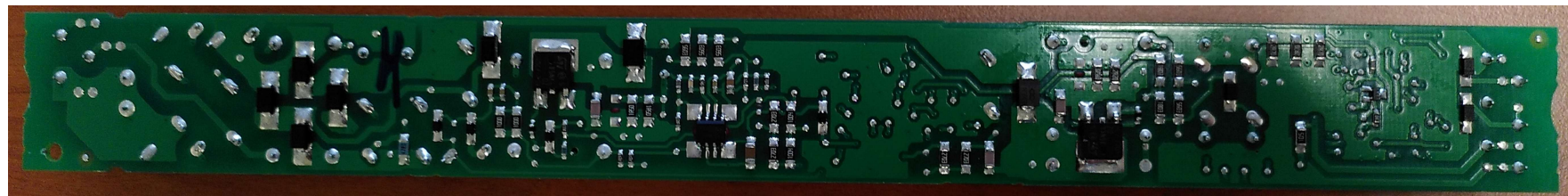
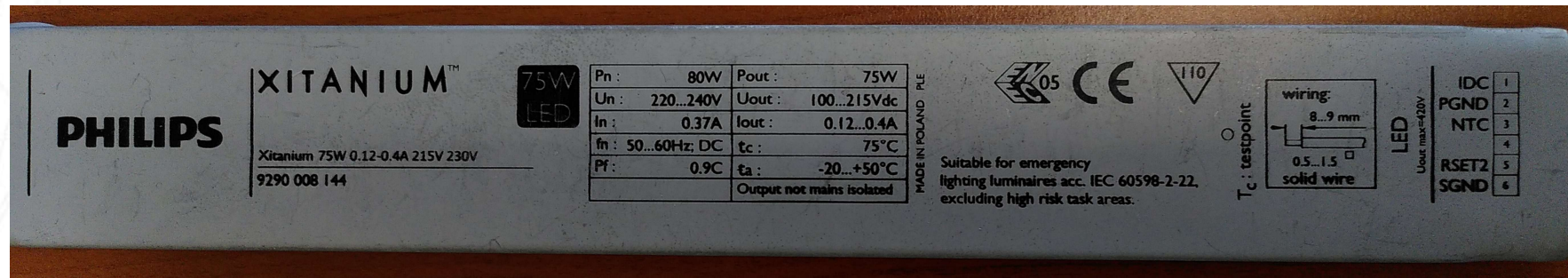
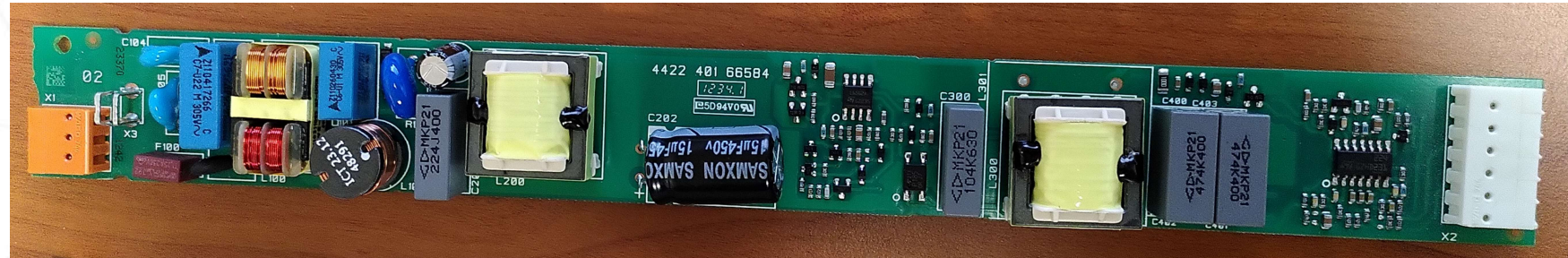
Introduction

Boost
Structure
PWM control
Hysteresis control

Flyback

Example of
converter

Example: LED driver



Introduction

Boost

Flyback

Example of
converter

Flyback for PFC

Introduction

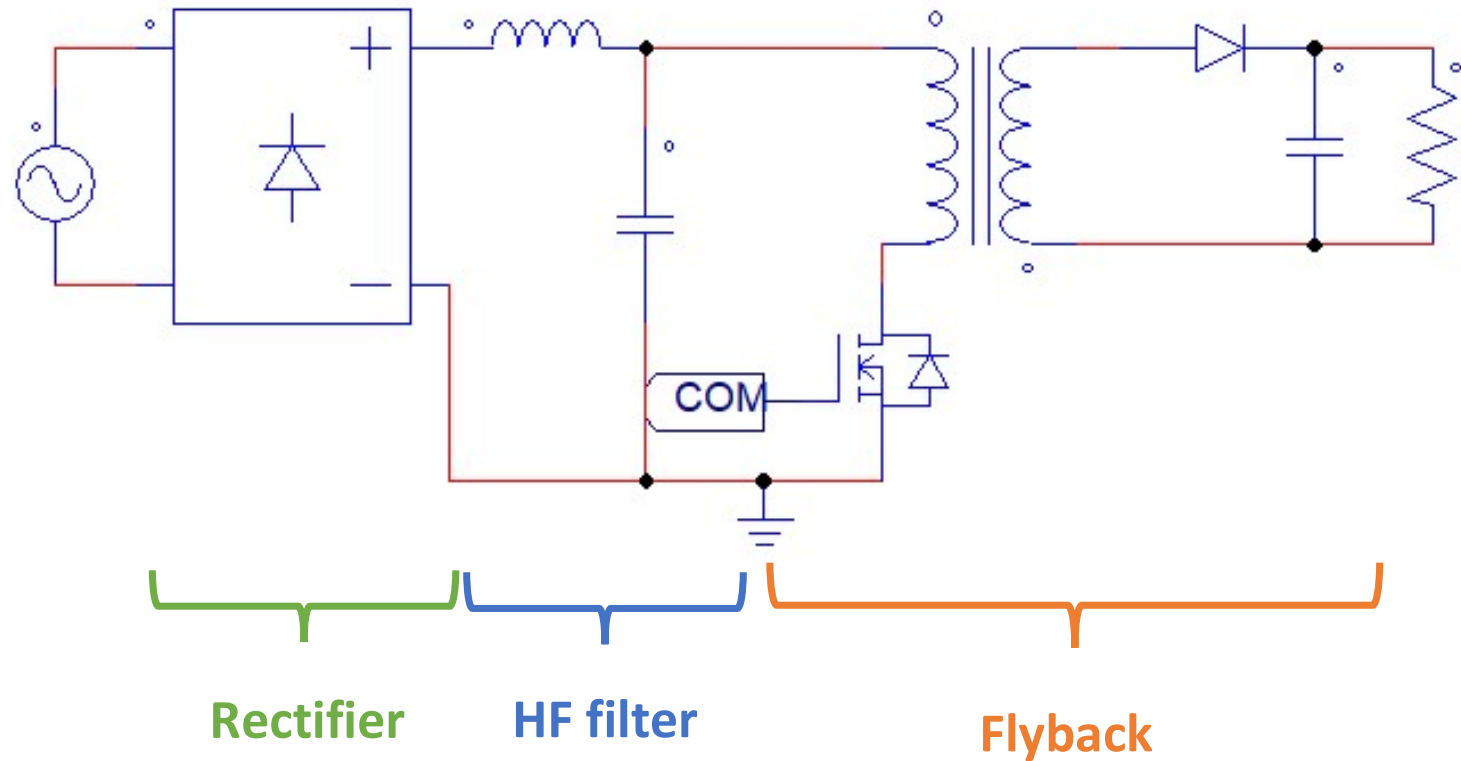
Boost

Flyback Structure

Example of RT7306
Control

Example of
converter

Structure



Introduction

Boost

Flyback

Structure

Example of RT7306

Control

Example of converter

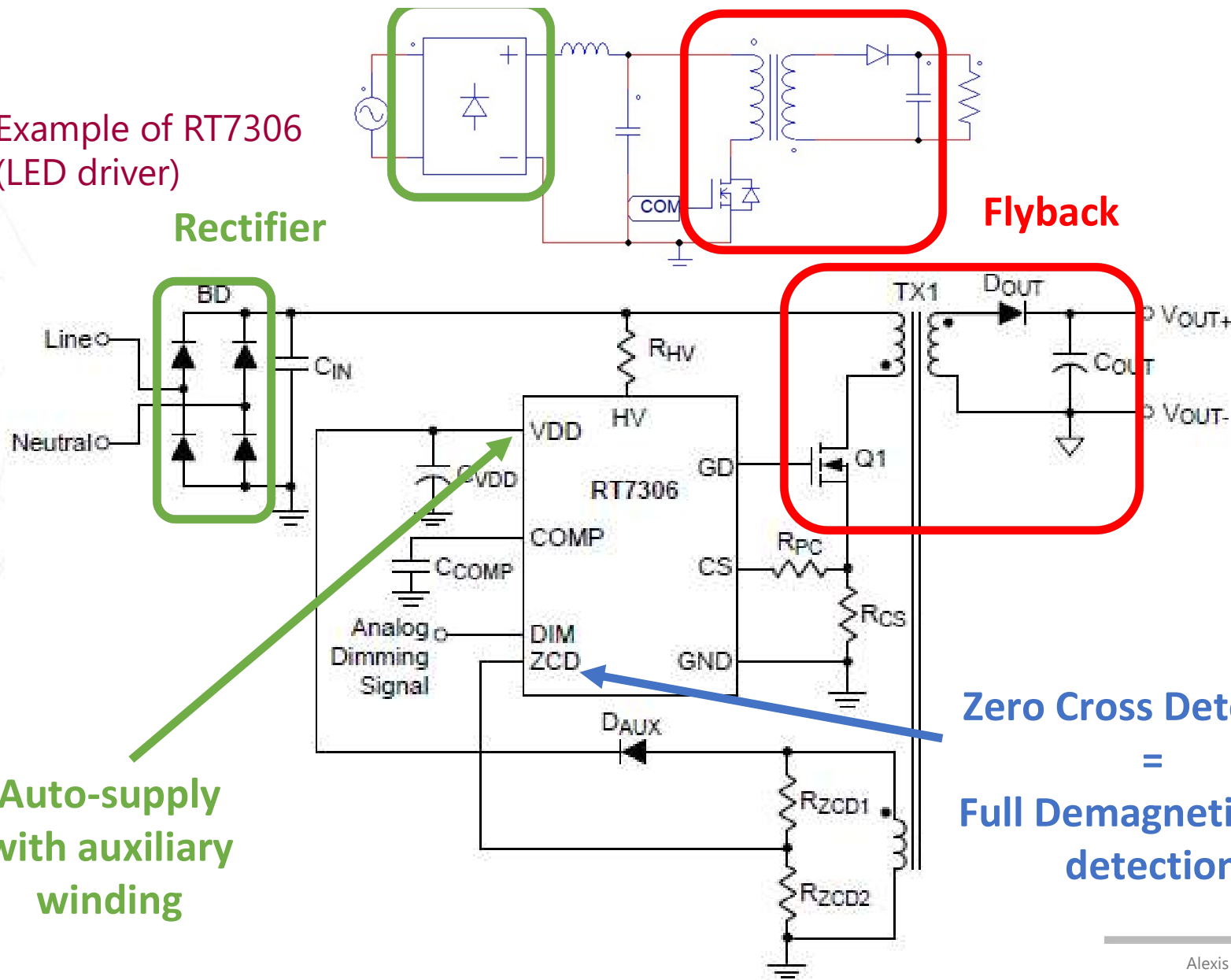
Example of RT7306
(LED driver)

Rectifier

Flyback

Auto-supply
with auxiliary
winding

Zero Cross Detection
=
Full Demagnetization
detection



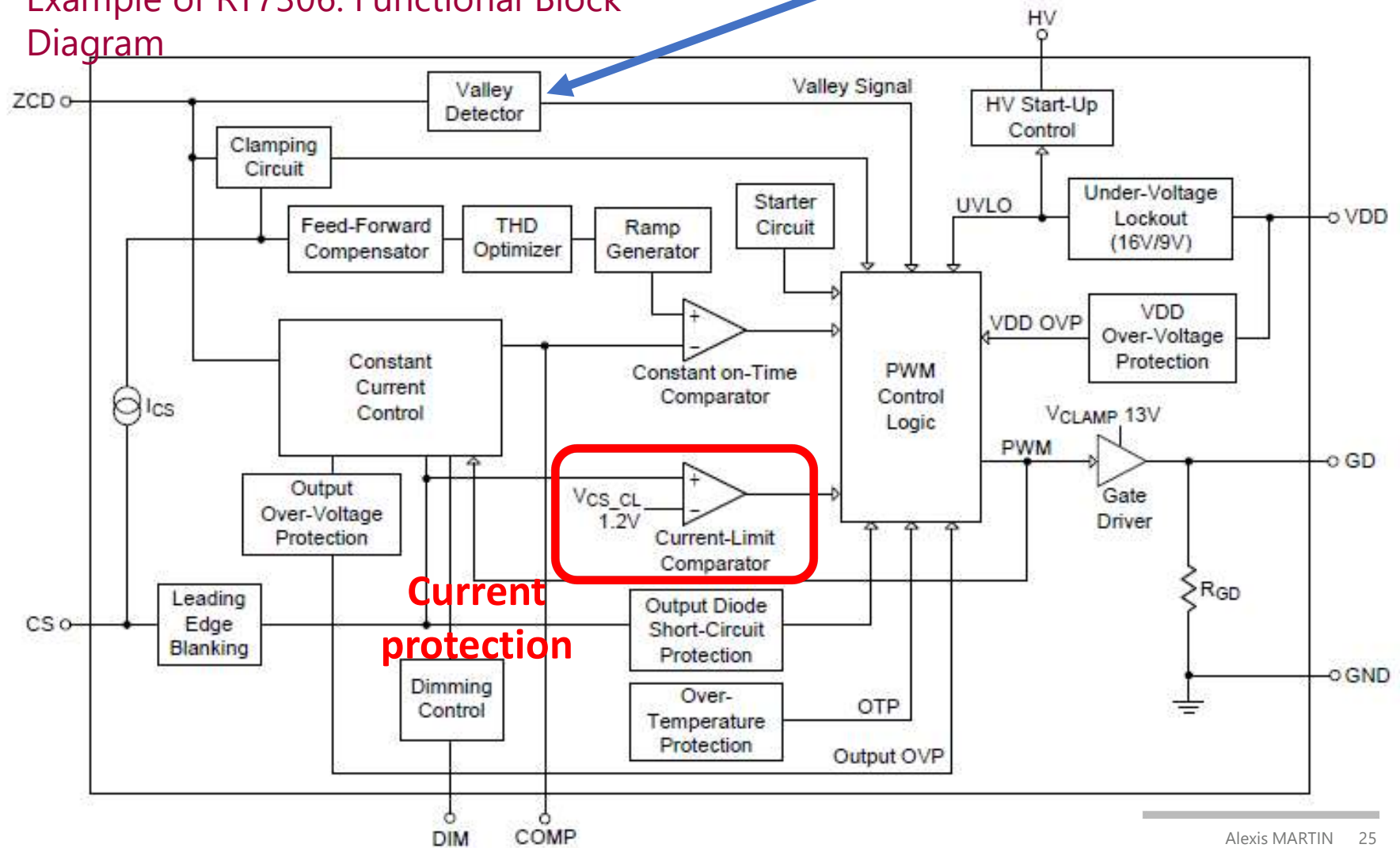
Introduction

Boost

Flyback
Structure
Example of RT7306
Control

Example of
converter

Example of RT7306: Functional Block Diagram



Full demagnetization
detection

Current
protection

Introduction

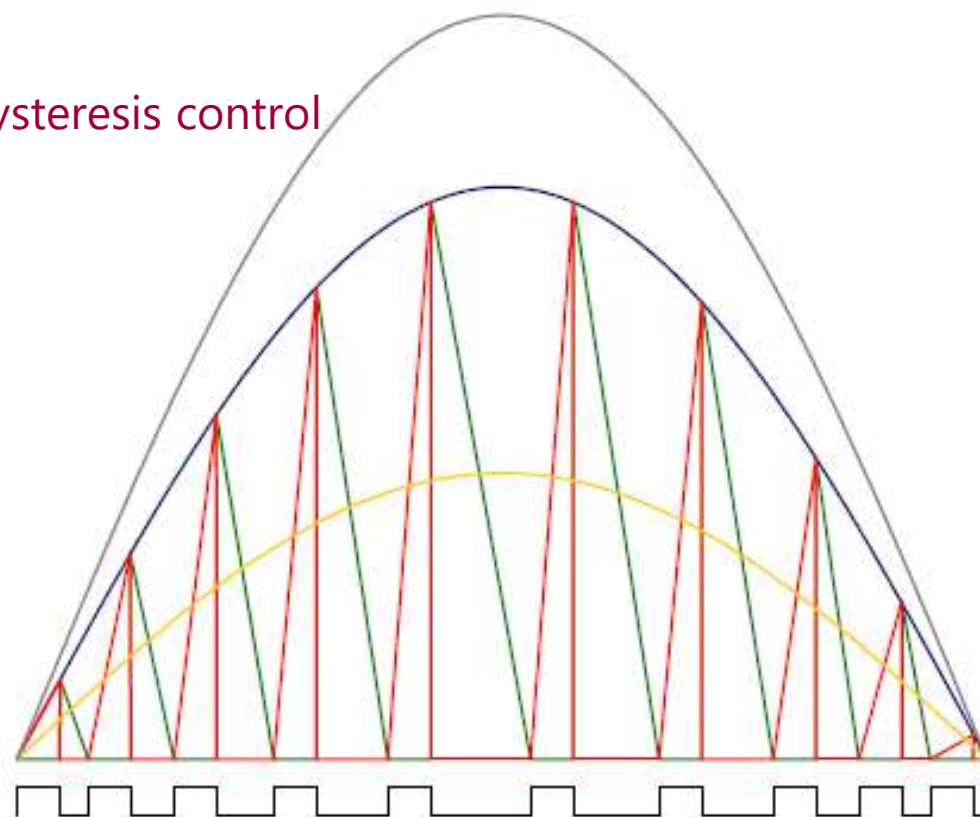
Boost

Flyback

Structure
Example of RT7306
Control

Example of
converter

Hysteresis control



- VIN Input Voltage
- IL_PK Peak Inductor Current
- IQ1_DS MOSFET Current
- Iin_avg Average Input Current
- IDo Output Diode Current
- VQ1_GS MOSFET Gate Voltage

$$I_{L_PK} = \frac{V_{IN}}{L_m} \times t_{ON}$$

$$I_{L_PK} = \frac{V_{IN_PK} \times |\sin(\theta)| \times t_{ON}}{L_m}$$

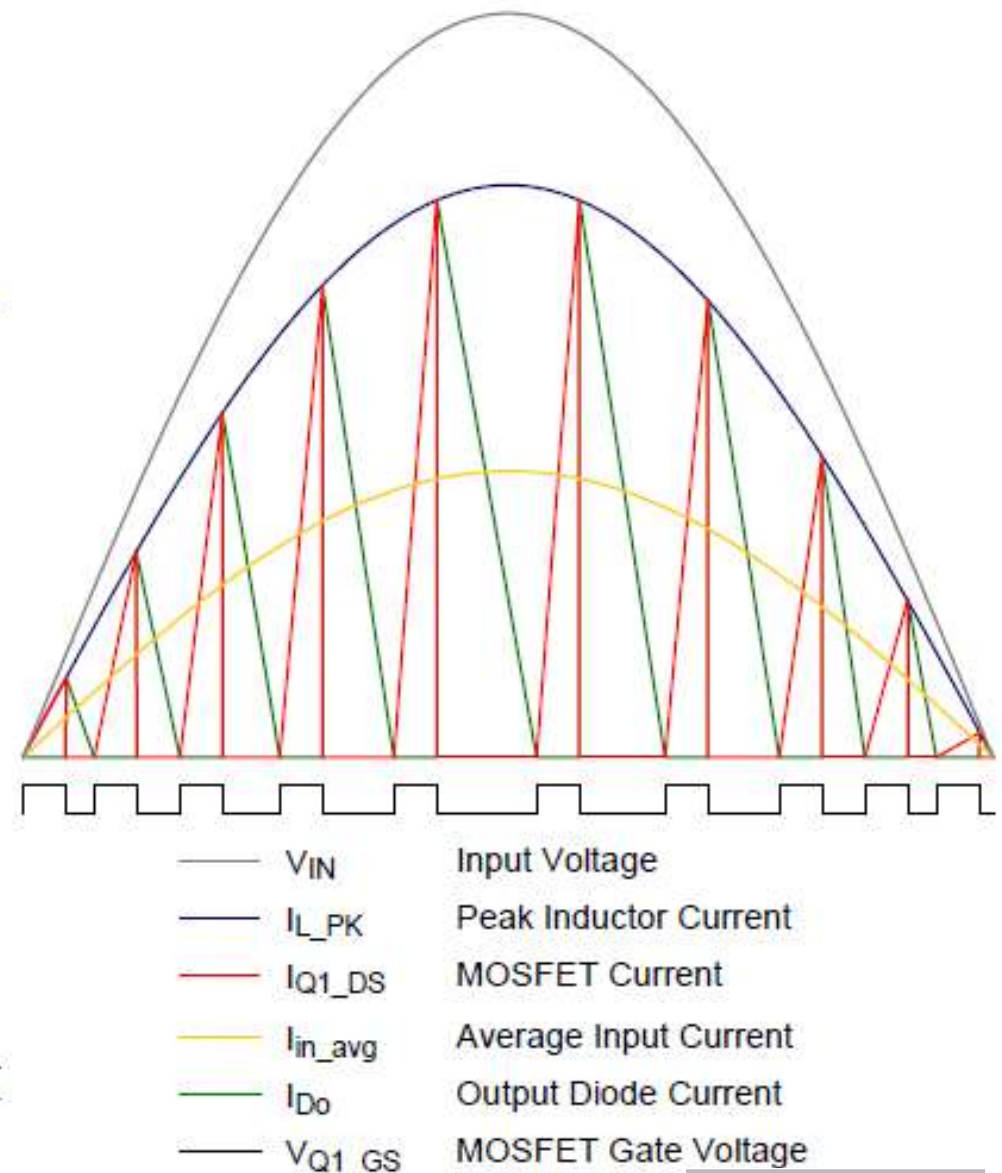
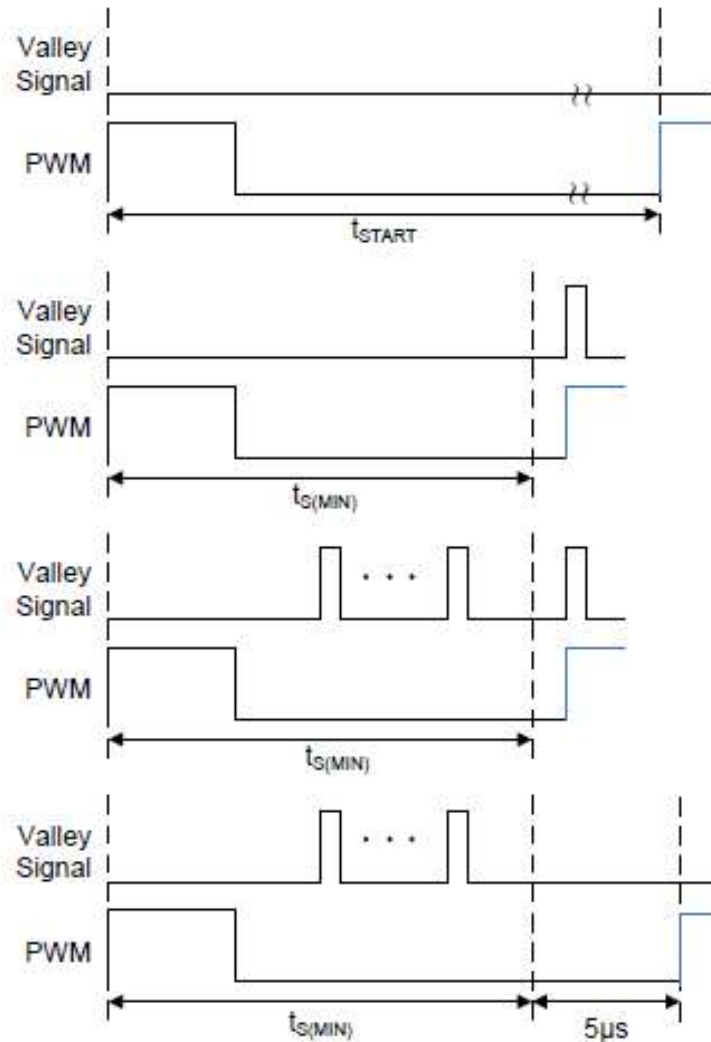
Introduction

Boost

Flyback Structure Example of RT7306 Control

Example of converter

Constant on-time control



Introduction

Boost

Flyback

**Example of
converter**

Example of a full converter

Introduction

Boost

Flyback

Example of
converter

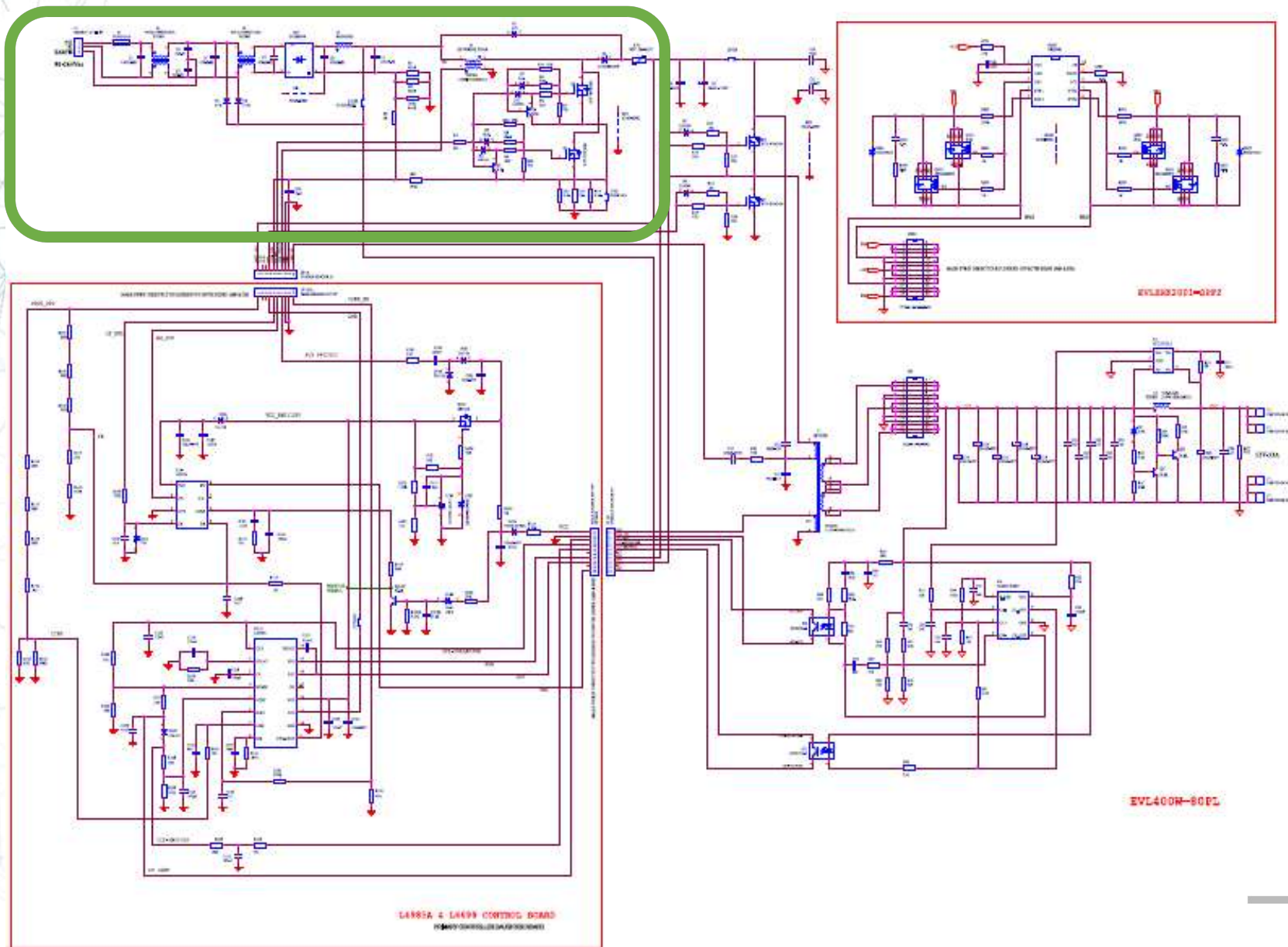
Example: EVL400W-80PL



Example: EVL400W-80PL

The architecture is based on a two-stage approach: a front-end PFC pre-regulator based on a CCM (Continuous Conduction Mode) boost PFC controller using the L4985A, and a downstream LLC resonant half-bridge converter, designed around the L6699.

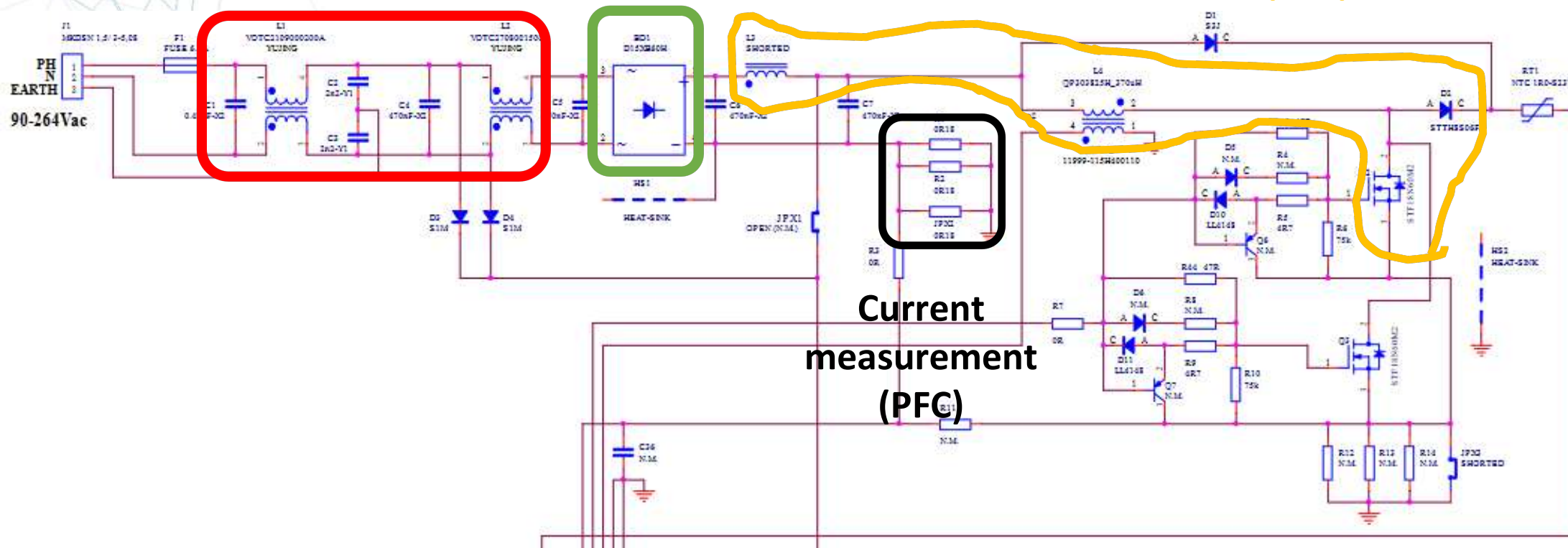
- Universal input mains voltage range: from 90 Vac to 264 Vac – frequency from 45 to 65 Hz
- Output voltage: 12 V at 33 A continuous operation
- Overall efficiency at full load: > 89%, according to ENERGY STAR® 6.1 limit for computer and compliant with 80Plus PLATINUM level
- Average efficiency: > 89%, according to European CoC ver. 5 Tier 2 for external power supplies
- Efficiency at 250 mW > 50%, compliant to EuP lot 6 Tier 2 limit for household and office equipment
- No load mains consumption: < 150 mW at 230 Vac, below European CoC ver. 5 Tier 2 limit for external power supplies
- Mains harmonics: meets EN-61000-3-2 Class-D and JEITA-MITI Class-D
- EMI: according to EN55022 Class-B
- Safety: meets EN60950 standards
- RoHS compliant

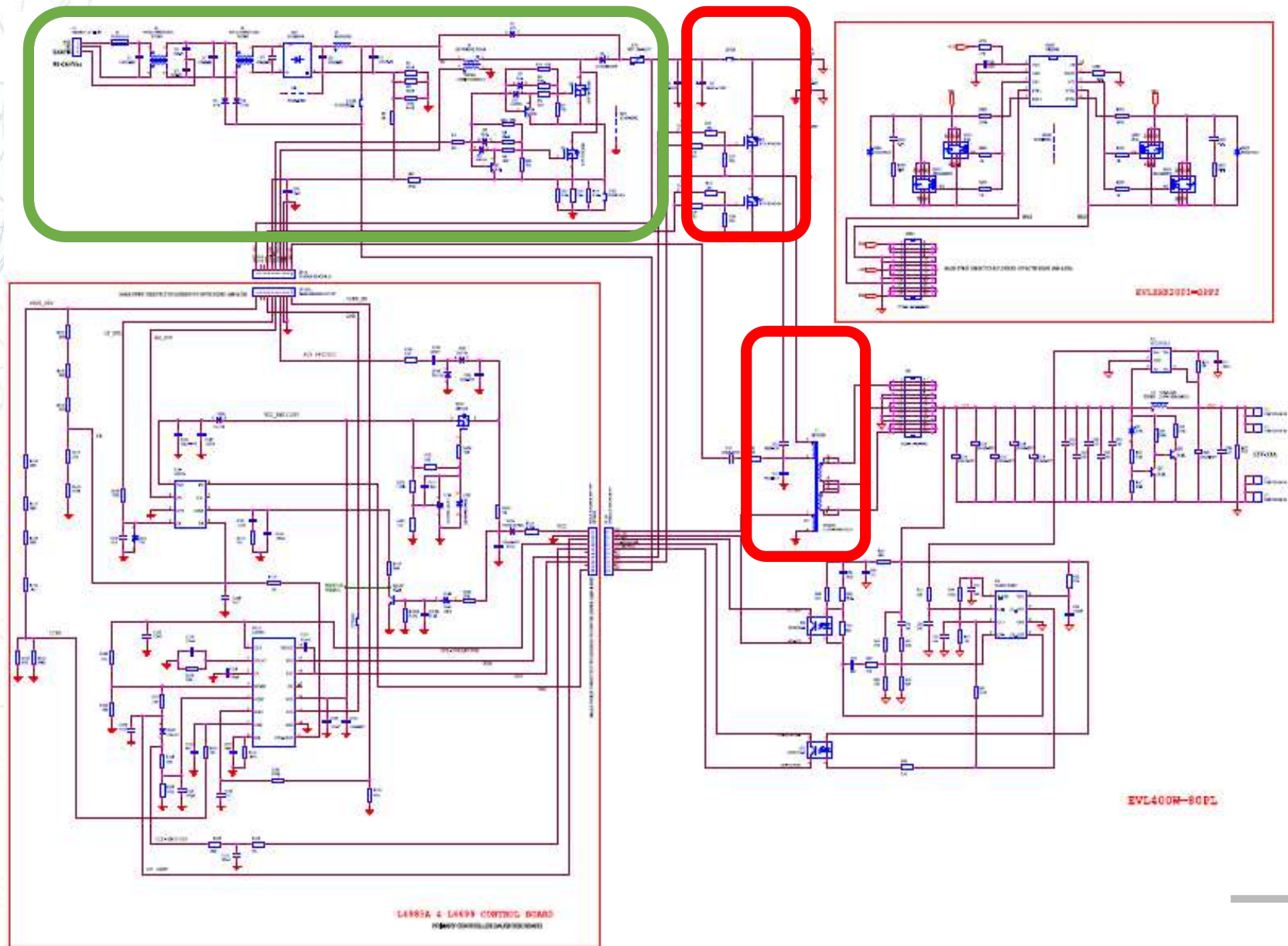


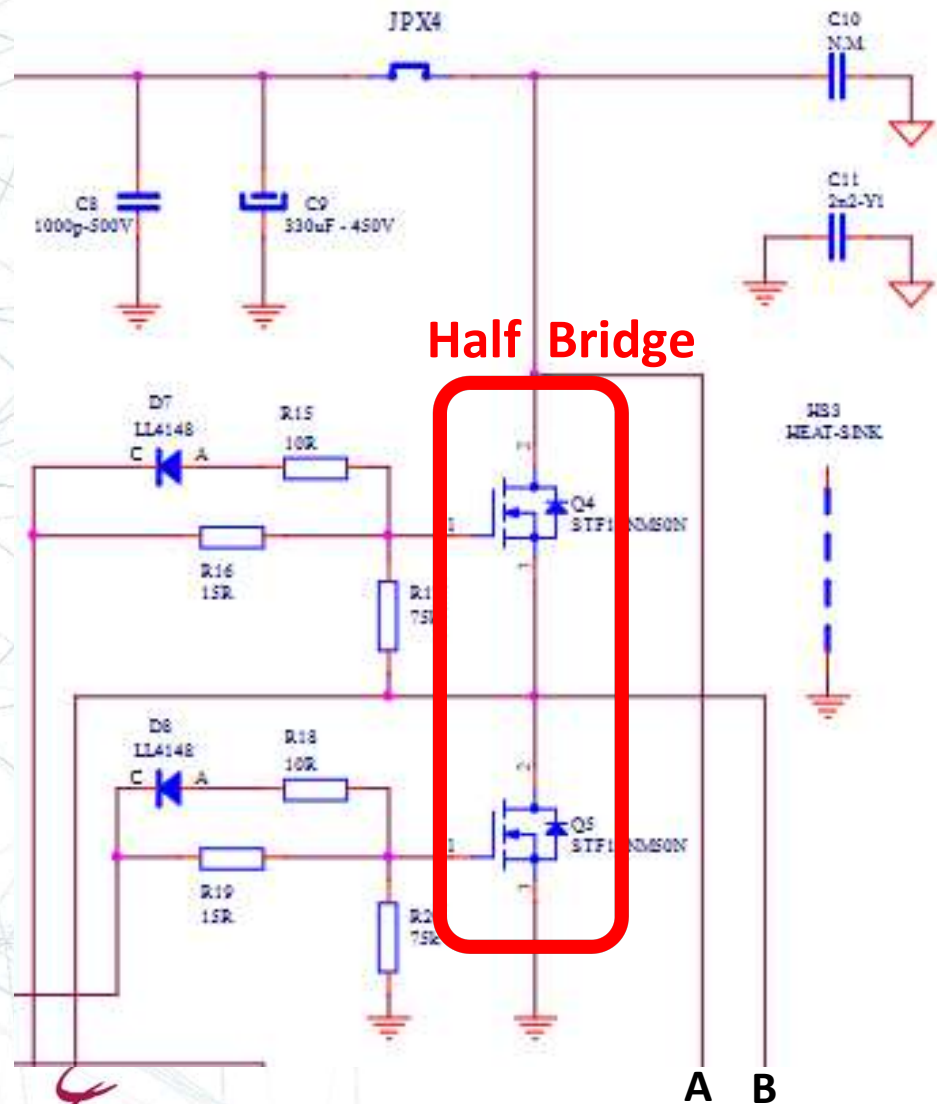
HF filtering

Rectifier

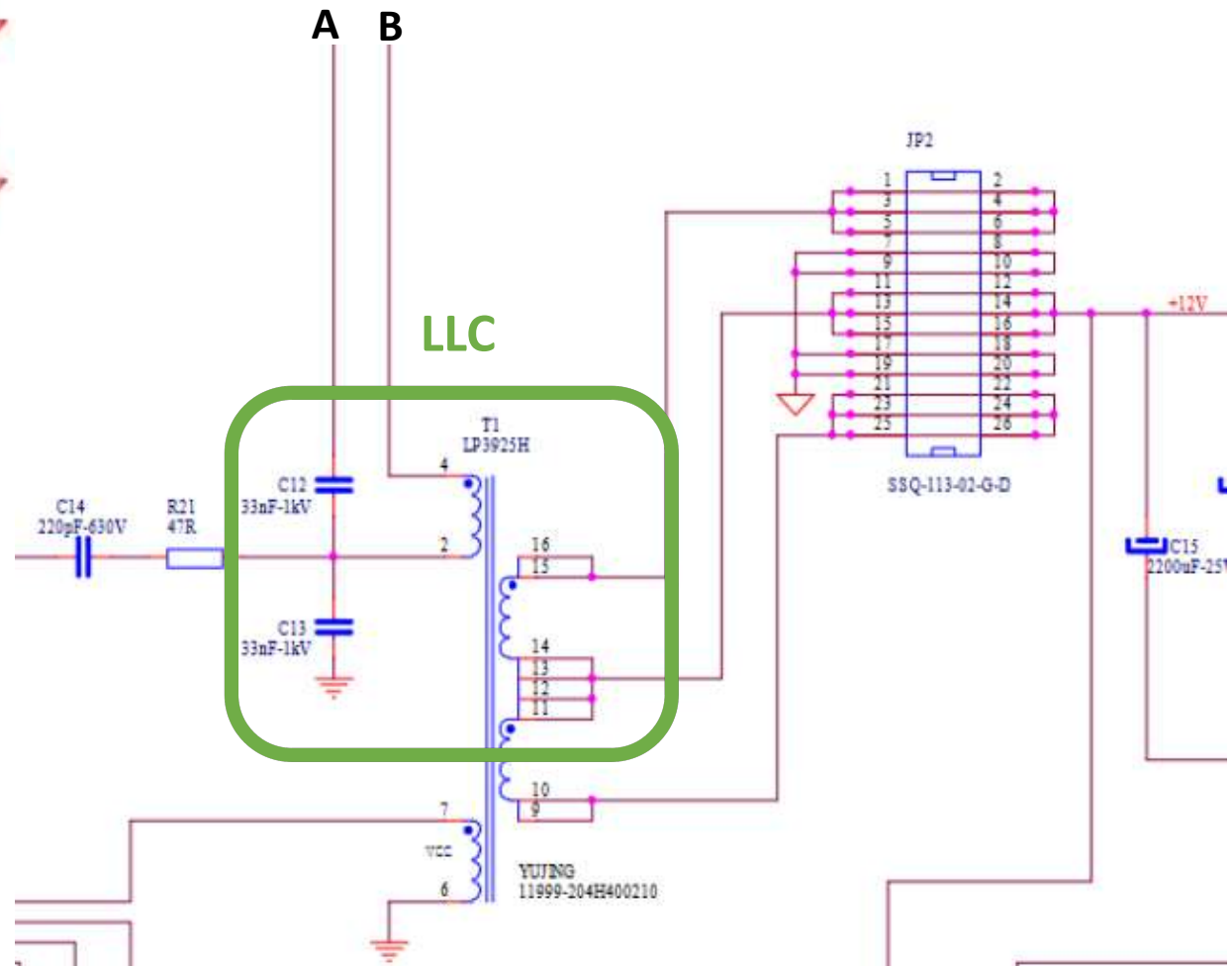
Boost (PFC)



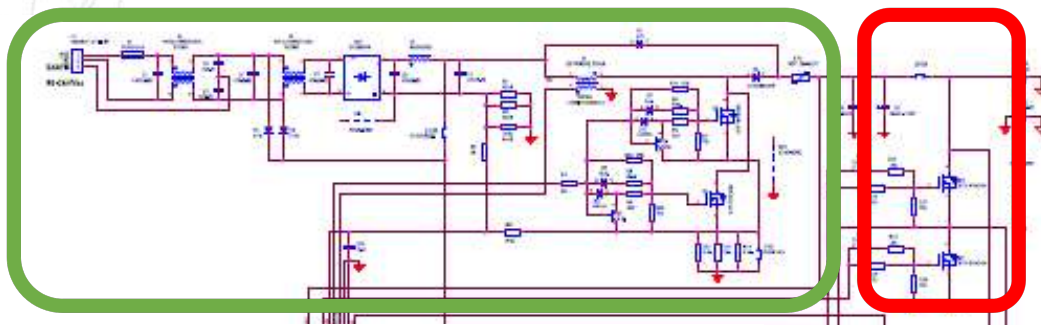




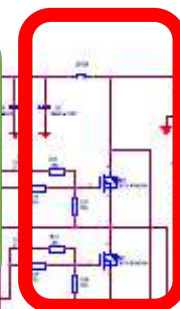
Half Bridge



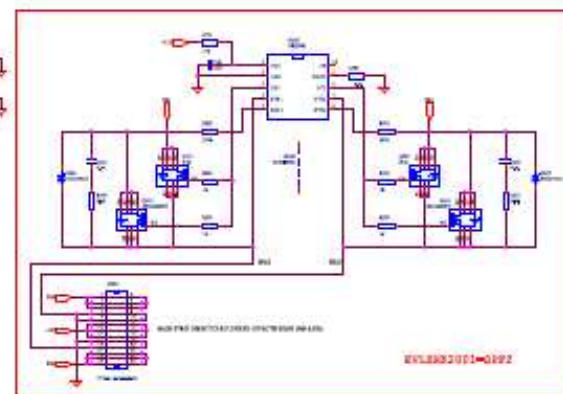
LLC



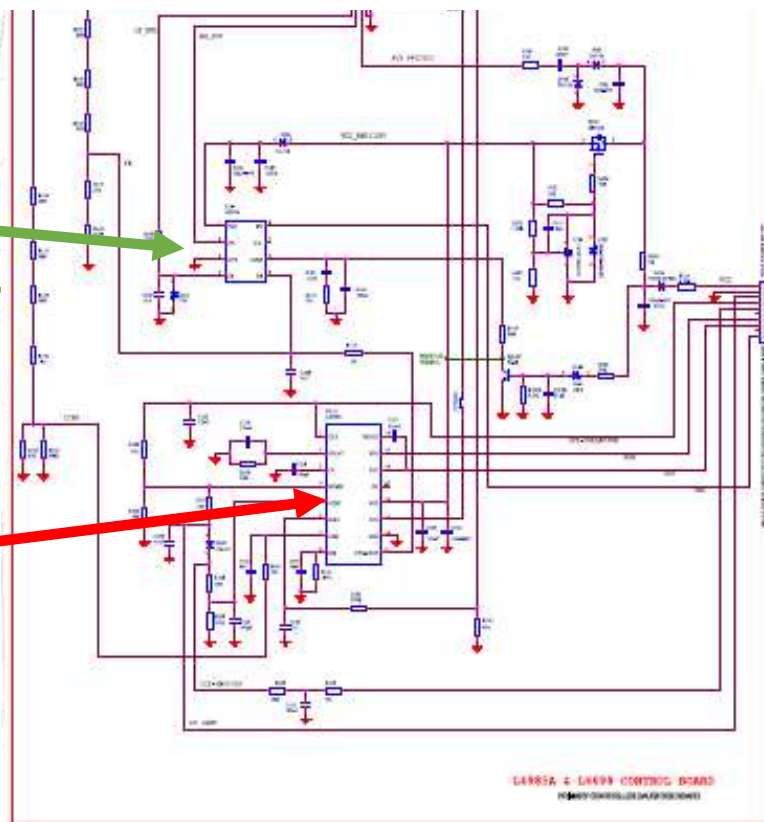
HF filtering + rectifier + Boost (PFC)



Half Bridge



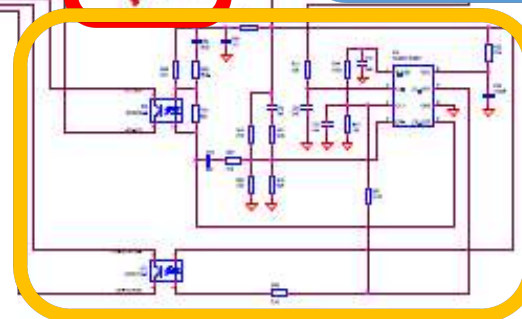
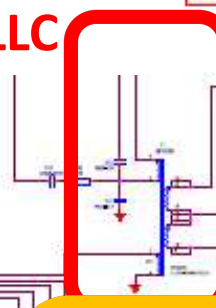
Synchronous rectifier (for LLC)



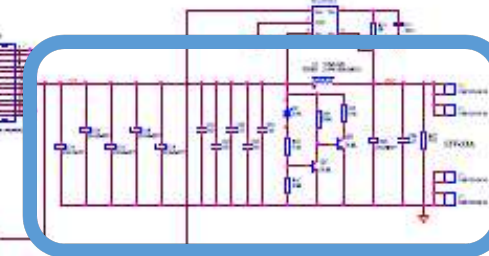
Boost (PFC) controller L4985A

Half Bridge LLC controller L6699D

LLC



Output measurement + optocoupler (isolation) for Half Bridge LLC



Output filtering



Example: EVL400W-80PL

2 transistors of Boost

**Synchronous
rectifier
(for LLC)**

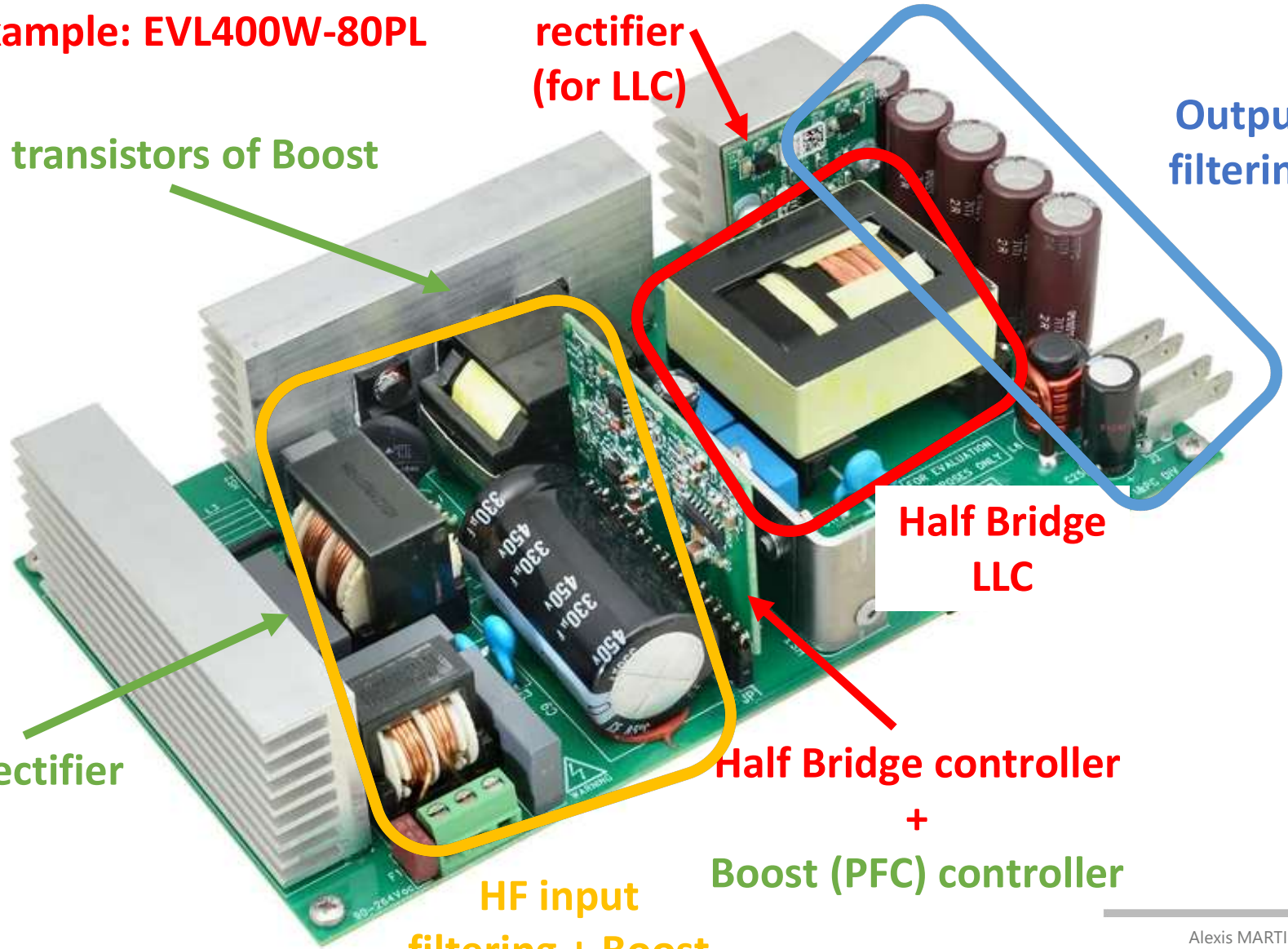
**Output
filtering**

**Half Bridge
LLC**

Rectifier

**Half Bridge controller
+
Boost (PFC) controller**

**HF input
filtering + Boost**



Example: EVL400W-80PL

